

# INSTITUTE OF ACTUARIES OF INDIA

## EXAMINATIONS

**17<sup>th</sup> March 2017**

**Subject ST2 – Life Insurance**

**Time allowed: Three hours (14.45\* – 18.00 Hours)**

**Total Marks: 100**

### INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions inside the cover page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
2. *\* You have 15 minutes at the start of the examination in which you are required to read the questions. You are strongly encouraged to use this time for reading only, but notes may be made. You then have three hours to complete the paper.*
3. *You must not start writing your answers in the answer sheet until instructed to do so by the supervisor.*
4. *The answers are not expected to be any country or jurisdiction specific. However, if Examples/illustrations are required for any answer, the country or jurisdiction from which they are drawn should be mentioned.*
5. *Attempt all questions, beginning your answer to each question on a separate sheet.*
6. *Mark allocations are shown in brackets.*
7. *Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

**AT THE END OF THE EXAMINATION**

**Please return your answer book and this question paper to the supervisor separately.**

**Q. 1)** A small proprietary life insurance company, ABC Life, which was incorporated two years ago, has been approached by a large competing financial services company XYZ Ltd that wishes to acquire ABC Life.

**i)** Define the appraisal value of a life insurance company. (1)

**ii)** Outline how the answer in part i) would apply to ABC Life in the context of the acquisition proposal (any steps in the calculation process need not be described). Comment on any particular components which are likely to be subjective. (5)

ABC Life has sold only with profits business in its two years of operations, and has not modeled asset shares very accurately so far. It is considering setting up a sophisticated framework to track asset shares.

**iii)** Discuss why accuracy in the asset share calculations is necessary for ABC Life both to complete the potential acquisition and also for its ongoing management. (12)

**iv)** Define asset share and list all the components (with definitions where relevant) that need to be considered as part of the calculations. (6)  
[24]

**Q. 2)** A large life insurance company has been selling a wide range of savings and protection products for many years, with a significant portion of the product portfolio being unit-linked insurance.

The company's marketing department has proposed that the company should sell only unit-linked insurance in future, since these contracts pose minimal risk to the company.

Discuss the risks to the company from the unit-linked business compared to other products and indicate how these risks may be mitigated.

[10]

**Q. 3)** A life insurance company sells both conventional and accumulating with profits endowment and whole of life products.

**i)** Explain why the company would have a smoothing policy as part of its with profits management framework, and discuss the items that such a policy would cover. (7)

**ii)** Describe the overall principles to be followed for determining the terminal bonus for policies in the coming year, mentioning any special considerations required for accumulating with profits business. (5)

**iii)** Outline the calculation process to derive the terminal bonuses based on these principles. (3)

The company recently declared its terminal bonus rates, but within the last few weeks since the declaration, there has been a large fall in the domestic stock market.

- iv) Discuss the risks arising from the recent economic environment and the issues the company should consider when revising the terminal bonus scales. (10)  
[25]

**Q. 4)** For its large block of unit-linked business, a life insurance company internally manages the funds and calculates unit prices. As part of the unit pricing, the company derives the appropriation and expropriation prices for units.

- i) Compare and contrast appropriation and expropriation prices. (2)
- ii) Briefly discuss what the company needs to do to ensure the unit pricing meets the requirements of the “basic equity principle”. In practice, can the company make some simplification to this process? (6)  
[8]

**Q. 5)** A life insurance company has been selling the following non-profit deferred annuity product for the last 15 years to lives aged 45 years. The details are as follows:

*During the deferred period:*

|                  |                                     |
|------------------|-------------------------------------|
| Premium:         | Single premium                      |
| Death benefit:   | Equal to 100% of the single premium |
| Surrender Value: | Equal to 95% of the single premium  |

The policy would vest when the policyholder turned 60 years of age. At the time of vesting of the annuity, the policyholder has the option to either take lump-sum benefit of  $k\%$  of the single premium, or get the following annuity benefit starting from age 60 till the age he/she survives:

|                          |                                    |
|--------------------------|------------------------------------|
| Annuity amount (annual): | Equal to 10% of the single premium |
| Death benefit:           | nil                                |
| Surrender benefit:       | nil                                |

The lump sum alternative to the annuity is not guaranteed, but product literature issued throughout the last 15 years has shown figures assuming  $K$  is 175. So far, none of the policies have entered the annuity payment phase. However, 10% of the in force policies by number are expected to enter the annuity payment phase next year.

- i) List the assumptions, other than investment assumptions, that would be required while calculating the statutory reserves for this product this year using a deterministic method. (5)

During the analysis of surplus exercise undertaken at the end of the following year once retirements have started, it is observed that a large proportion of the policyholders opted to take 175% of the single premium instead of continuing the policy, and that this is reducing the company's embedded value. On further analysis, it was found that 175% was too high a discontinuance benefit to provide at the time of vesting.

A senior colleague has commented that this discontinuance benefit is, in fact, like a surrender benefit at the time of vesting of the policy.

- ii) Based on this suggestion, state the principles that should be considered while deriving the appropriate level of discontinuance benefit. (4)
- iii) Discuss which of the above parameters the current definition of the discontinuance benefit does not satisfy. (4)
- iv) Describe how the appropriate level of surrender value to be offered at the time of vesting of the benefit may be derived using the prospective method including the high level calculation (and working assumptions) that would be required. (8)

It has been proposed that this opportunity be used to redesign the product. The changes proposed include: introducing a surrender benefit and a death benefit post the vesting of the annuity, with the aim of retaining customers past the date of vesting.

- v) Discuss the factors to be considered while evaluating a new product design, with reference to this proposal. (12)
- [33]**

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